

Shaan Pakala

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About

I am a 2nd year Computer Science Ph.D. student at the University of California, Riverside, where I work on AI4Science research problems with my advisor Professor [Vagelis Papalexakis](#). This includes developing interpretable surrogate models for designing new materials, or efficiently generating expensive physics simulation data using generative machine learning.

Education

University of California, Riverside

- **Ph.D. in Computer Science** **Sept. 2025 – Present**
- **B.S. in Data Science** **Sept. 2021 – June 2025**

Research Experience

Lawrence Livermore National Laboratory

- **Graduate Research Intern** **June 2026 – Present**
Intern in the predictive healthcare group, with Dr. Priyadip Ray. I work on using machine learning for early disease outbreak detection, using the Tuolumne supercomputer (#14 on [TOP500](#) list).
- **Graduate Research Intern** **June 2025 – Sept. 2025**
Intern in the predictive healthcare group, with Dr. Priyadip Ray and Dr. Braden Soper. I worked on causal inference for Parkinson's Disease progression, with clinicians/biomedical researchers.

University of California, Riverside

- **Graduate Student Researcher** **June 2025 – Present**
I work with domain-experts in physics & material science on AI4Science problems using tensor decomposition: (i) surrogate modeling for robust & interpretable material design [1] [3] and (ii) efficient physics simulation generation using tensor decomposition in GANs & diffusion models [4].
- **Undergraduate Student Researcher** **June 2024 – June 2025**
Research with Prof. Papalexakis. I worked on automating the optimization of data science pipelines (e.g., neural network architecture search, SQL queries) with tensor completion [2].

Papers (** denotes equal contribution*)

Conference

- [1] **Tensor Methods: A Unified and Interpretable Approach for Material Design**
[Shaan Pakala](#), A.E. Gongora, B. Giera, and E.E. Papalexakis
ACM SIGKDD (AI for Sciences track) 2026 [PDF] [Code]
- [2] **Automating Data Science Pipelines with Tensor Completion**
[Shaan Pakala](#), B. Graw, D. Ahn, T. Dinh, M.T. Mahin, V. Tsotras, J. Chen, and E.E. Papalexakis
IEEE International Conference on Big Data 2024 [Link] [PDF] [Code]

Workshop

- [3] **Surrogate Modeling for the Design of Optimal Lattice Structures using Tensor Completion**
[Shaan Pakala](#), A.E. Gongora, B. Giera, and E.E. Papalexakis
NeurIPS AI for Accelerated Material Design Workshop 2025 [PDF] [Code]
- [4] **Efficiently Generating Multidimensional Calorimeter Data with Tensor Decomposition Parameterization**
P. Goulart*, [Shaan Pakala](#)*, and E.E. Papalexakis
ICCV Representation Learning with Very Limited Resources Workshop 2025 [PDF] [Code]

Awards

- **Travel Awards** (3): ACM KDD'26, NeurIPS'25 AI4Mat Workshop, and IEEE BigData'24
- **Undergraduate Research Spotlight Award** 2025
Bourns College of Engineering – University of California, Riverside
- **NSF Research Experience for Undergraduates** 2024
University of California, Riverside

Contributed Talks & Posters

Talks

- **Tensor Methods for Material Design** 2026
Lawrence Livermore National Laboratory Annual [CASIS Workshop](#)
- **Surrogate Modeling with Tensor Completion** 2026
UTRGV [MECIS Spring Webinar](#)

Posters

- **Tensor Decomposition for Interpretable Exploration of Scientific Design Spaces** 2026
SIAM SDM Ph.D. Forum
- **Tensor Methods for Interferometer Simulations** 2026
CECI [Workshop on Detector Science](#)
- **Tensor Methods for Material Design** 2026
SAIR [Science & AI Summit](#)
- **Tensor Completion for Material Property Prediction** 2025
AAAI Bridge on Knowledge-guided Machine Learning

Academic Service

Organizer

- [AI4AutoSci](#) Workshop at IEEE BigData'26 2026

Reviewer

- TensorKDD Workshop at ACM KDD'26 2026
- AI Data Scientist Workshop at ACM KDD'26 2026